

Work

Tools Intern

May - August 2024

Infinity Ward

- Implemented UI using PyQt to represent level content in a spatial hierarchy, making it easy to find and select content.
- Used Model/View architecture in PyQt to separate data from graphics, improving maintainability and testability.
- Optimized level baking by batching calls to C++ engine code and multithreading Python code, reducing iteration time.
- Created UI layouts using XML to improve iteration time, readability, and long-term maintainability.
- Wrote automated tests in Python to extend coverage of continuous integration to new tools and features.
- Responded to bug reports by reproducing and fixing issues then following up with reporters, ensuring tool stability.
- Wrote technical documentation using Confluence and P4 CL descriptions to explain intended use and design choices.
- Collaborated with global engine and pipeline teams on exposing level data to my tool ensuring a responsive UI.
- Participated in code reviews via Swarm and applied feedback to improve the quality of new and existing code.

Projects

Tools & Gameplay Programmer

August 2025 - current

Wash Away

- Consolidated a workflow for editing multiple actors by creating a one-stop interface with Unity C# property drawers.
- Increased accuracy and speed of collision data editing by implementing a 2D collision algorithm and in-engine editor.
- Modified open source, C++/Qt drawing software *Krita* to support dual-layer editing, eliminating pipeline bottlenecks.
- Wrote XML files to format new interface options in *Krita*, reducing need for error-prone C++ setup.

Lead Game Designer

August - December 2024

Stumblebumps

- Prototyped concepts during a game jam, pitched to faculty, and got chosen to lead a 15-person capstone project.
- Coded a C# event/interface system to sync characters with moving, rotating platforms and produce expected physics.
- Debugged UI by analyzing another engineer's code, identifying an input routing error, and implementing a quick fix.
- Helped artists deliver game-ready assets on strict deadlines by creating video guides on workflows and best practice.
- Achieved highest honors at our university game awards thanks to the enthusiasm and dedication of the whole team.

Shader Programmer

April 2023

Splatton Recreation

- Used HLSL, render textures, and shader graphs to write, store, and render paint that appears similar to *Splatton*.
- Optimized GPU render texture readback by moving to an asynchronous solution which improved scalability.
- Implemented character physics in C# from scratch to support readback of render textures and six movement states.

Skills

Technologies: C++, C#, Python, Qt, Unreal Engine, Unity, Git, Perforce, Visual Studio, XML

Core Competencies: Tools, User Interface, Learning Unfamiliar Code, Debugging, Math (linear algebra, calculus, discrete)

Education

B.S. Computer Animation and Game Development
California State University Chico. Dec. 2024

A.S. Computer Science
Bakersfield College. May 2022.